Purwadhika Digital Technology School | Digital For Executive

Data Analysis

Session 10: Statistics II

Digital for Executive

Outline

- Association
- Probability
- Inferential Statistics : Hypothesis Testing

Association

Association

Association

Type of Association based on variable

How To Explore ?

Exploring Association

- Smoking status and 20 year survival in women
- Education and crime rate
- Death penalty and race

Relationship between two events

Two event often related to each other

for example:

- air temperature and humidity
- price and demand
- fertilizer and plant height
- weight and height

There are two types of relationship

- association → correlation
- causation → regression

Response variable and Explanatory Variable

For causation, when analyzing relationship between two variable usually we must first distinguish between **response variable** (y) and **explanatory variable** (x). In association we don't need that.

Causation. If change in X cause change in Y, doesn't imply that change in Y cause change in X. Ex, javanes (X) must be an indonesian but an indonesian (Y) could be sundaness, batak people, aceh people etc.

Response Variable:

- Value in response variable depends on explanatory variable.

Explanatory Variable:

- quantitative : How different value in explanatory relate to changes in response variable
- qualitative : it is like grouping or aggregating. how is the comparison of Y between group (based on some aggregate function such as mean, sum, count, percentage, etc)

Type of Cases Can happen in Relationship

Variable can be qualitative or quantitative. So, there are three possible cases:

1. qualitative vs qualitative:

   ex. gender and education
2. quantitative vs qualitative:

   ex. income and race, height and gender
3. quantitative vs quantitative:

   ex. air temperature and humidity, weight and height

How to explore the relationship ?

Qualitative vs Qualitative

- Graphical Summary: Barchart, Pie chart
- Numerical/Table Summary : Contingency table/cross tabulation, Odds ratio, Difference of proportion, Ratio of proportions, Chi-square Test.

Qualitative vs Quantitative

- Graphical Summary: Barplot
- Numerical/Table Summary: Aggregated table, Anova F-Test.

Quantitative vs Quantitative (we will focus on this)

- Graphical Summary: Scatterplot
- Numerical/Table Summary: Pearson Correlation or Spearman Correlation.

Correlation

- Correlation is about association and **association doesn't imply causation**.
- Correlation **doesn't differentiate response** (x) variable and **explanatory** variable (y).
- Correlation only **measure how strong relationship** and **the direction of relationship**.
- We will focus on numerical variable only

- Correlation ranged by

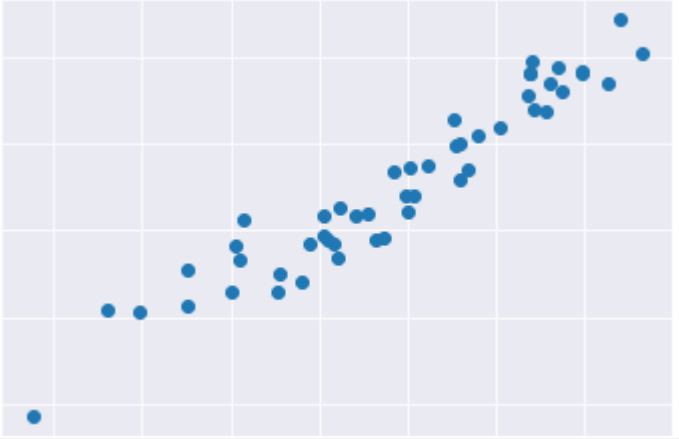
  $-1 < r < 1$
- no correlation (0)
- positive direction (+)
- negative direction (-)
- the magnitude (absolute value)

  0 - 0.3 : weak

  0.3 - 0.7 : medium

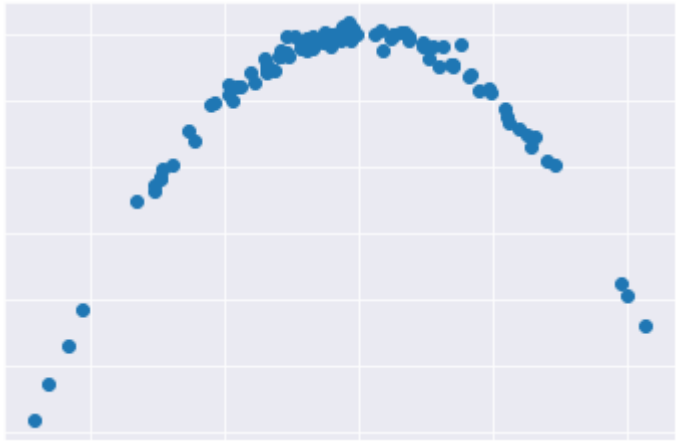
  0.7 - 1 : strong

Type of Numerical Relationship



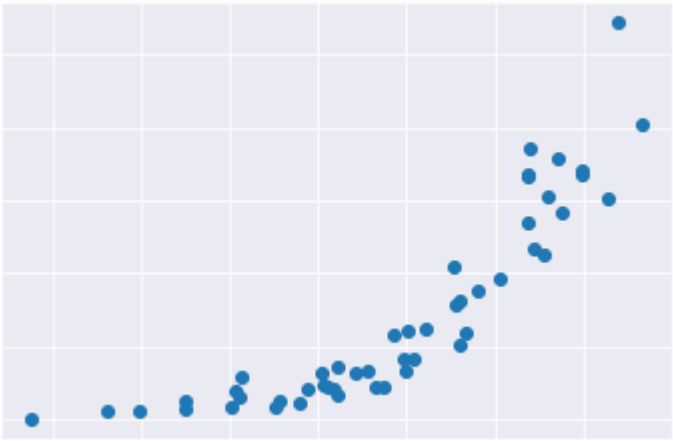
Linear:
Use Pearson Correlation

Ex. height and weight



Non Linear and Non-monotonic :
Strongly not recommended to measured by pearson or spearman.

Ex. fertilizer dose and plant height



Non-Linear or Monotonic:
Use Spearman Correlation

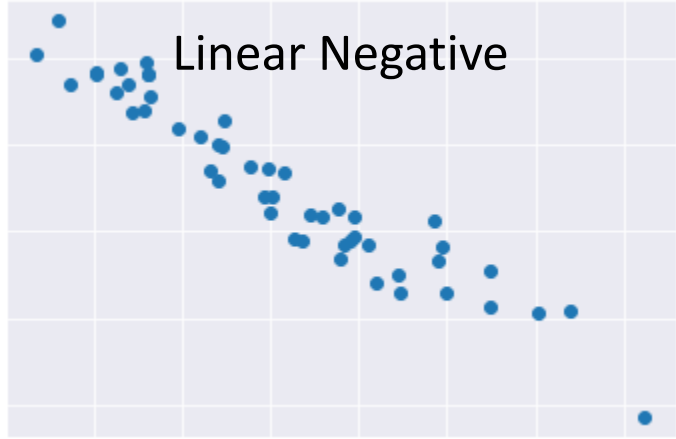
Ex. daily case of COVID-19

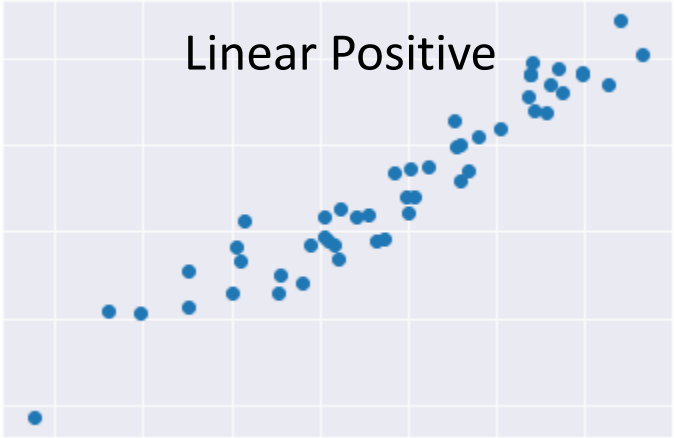
Pearson Correlation

1. Both of the variable should be quantitative
2. Relationship between two variable should be linear
3. Parametric method.

Formula

$$\rho_{X,Y} = \frac{E[(X-E[X])(Y-E[Y])]}{\sigma_X \sigma_Y}$$



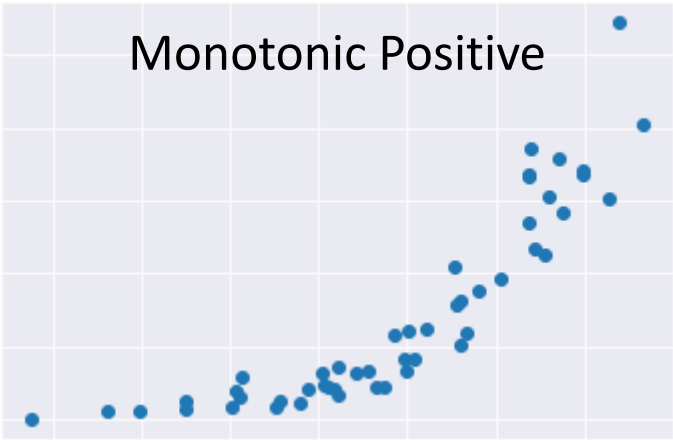


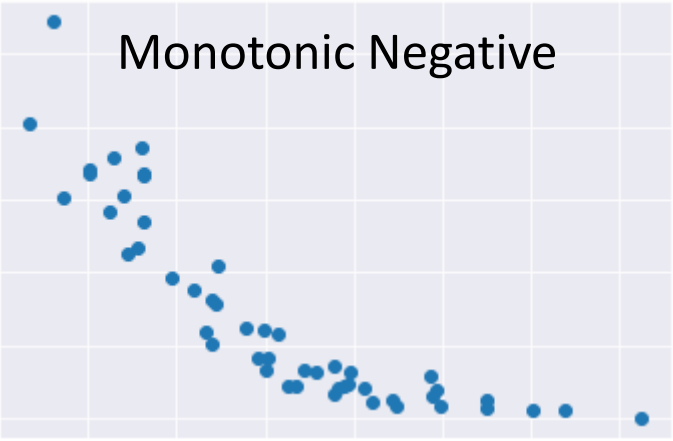
Spearman Correlation

1. Beside quantitative variable. it can be used to explore variable with ordinal scale.
2. Relationship between two variable should not be linear. It should be either positive monotonic or negative monotonic
3. Nonparametric version of pearson

Formula

$$\rho_{rank_x}, \rho_{rank_y} = \frac{cov(rank_x, rank_y)}{\sigma_{rank_x}, \sigma_{rank_y}}$$





Smoking Status and 20-year survival in Women

a survey of 1,314 women in the United Kingdom that asked each woman whether she was a smoker. Twenty years later, a follow-up survey observed whether each woman was dead or still alive

	Survival Status		
Smoker	**Dead**	**Alive**	**Total**
Yes	139	443	582
No	230	502	732
Total	**369**	**945**	**1,314**

- 31 % non-smoker died and 24% smoker died
- Smoker has lower death rate

Smoking Status and 20-year survival in Women

	Age Group							
	18–34 Survival?		35–54 Survival?		55–64 Survival?		65 + Survival?	
Smoker	Dead	Alive	Dead	Alive	Dead	Alive	Dead	Alive
Yes	5	174	41	198	51	64	42	7
No	6	213	19	180	40	81	165	28

	Age Group			
Smoker	18–34	35–54	55–64	65 +
Yes	2.8%	17.2%	44.3%	85.7%
No	2.7%	9.5%	33.1%	85.5%
Difference	0.1%	7.7%	11.2%	0.2%

For instance, for smokers of age 18–34, from The First Table the proportion who died was 5/(5 + 1742) = 0.028, or 2.8%

percentage of survival rate is vary for each age group

non-smoker always has lower death rate when age group taken into account

the association very different than before and more make sense

A phenomenon like this called Simpson's Paradox.

in which a trend appears in several different groups of data (partial) but disappears or reverses when these groups are combined (marginal)

Beware of the Simpson's Paradox when analyzing relationship : Education and Crime Rate

Education and Crime Rate

Urban Counties		Rural Counties	
Education	Crime Rate	Education	Crime Rate
70	140	55	50
75	120	58	40
80	110	60	30
85	105	65	25

Let's Analyze this data:

1. make the dataframe in python for whole dataset
2. analyze marginally
3. analyze partially

Beware of the Simpson's Paradox when analyzing relationship : Death Penalty and Race

Death Penalty and Race

Death Penalty, by Defendant's Race and Victim's Race				
Victim's Race	Defendant's Race	Death Penalty		
		Yes	No	Total
White	White	53	414	**467**
	Black	11	37	**48**
Black	White	0	16	**16**
	Black	4	139	**143**

Let's Analyze this data:

1. First, consider only those cases in which the victim was white. Find the conditional proportions that got the death penalty when the defendant was white and when the defendant was black. Describe the association.
2. Repeat part a for cases in which the victim was black. Interpret.
3. Repeat part a for cases in which we ignore the Victim's race

Probability

Probability

What is probability?

Probability distribution

- discrete
- continuous

Sampling Distribution

Central Limit Theorem

Probability

Probability study about randomness.

Many aspect in Statistics involve randomness (sampling from observation or assign subjects to treatments, assumption in hypothesis testing)

- When we roll a dice we know every possible outcome
  - 1, 2, 3, 4, 5, or 6
- but we don't know exactly what will occur

The proportion of any outcome (1, 2, 3, 4, 5, or 6) if we roll dice as much as possible. That is an example of probability.

P(any event) ranged from 0 to 1

Rolling Dice

- if the dice is balance the probability of each number occur should be ⅙.
- there is a way to estimate the probability by experimenting. we roll dice 60 times and record what is the record. We expect each number show up 10 times. But what we got is

	1	2	3	4	5	6
Throws	12	10	8	14	5	11
Relative Frequency	$\frac{12}{60}$	$\frac{10}{60}$	$\frac{8}{60}$	$\frac{14}{60}$	$\frac{5}{60}$	$\frac{11}{60}$

By looking at the result:

1. does it mean that the probability of each number occur is not equal to ⅙ ?
2. is it surprising we got one of the dice number 5 times ?

How about compute mean from any population

Let’s say we want to take sample randomly (30) from a purwadhika student population and compute the mean of the age

- Person 1 take sample and the result is 25.4
- Person 2 take sample again, are you sure the result is 25.4 again ?
- How about the next person ?

There are so many possibilities. You can say that the possibility is infinite if the population is very large.

Some Terminology in Probability

When rolling a dice the possible outcome are 1,2,3,4,5, or 6.

When rolling two dice the possible outcome are (1,1),(1,2),(1,3),...,(5,6), (6,6).

Every possible outcome is called **sample space**.

Subset of sample space is called **Event**.

- odd number (1,3,5)
- prime number (2,3,5)
- etc

Probability of an Event

The probability of an event A, denoted by P(A), is obtained by adding the probabilities of the individual outcomes in the event.

- When all the possible outcomes are equally likely,

$$P(A) = \frac{\text{number of outcomes in event A}}{\text{number of outcomes in the sample space}}$$

For example:
event A = odd number occur (1, 3, 5)

P(A) = 3/6

Probability Distribution

Probability distribution is a way to summarize every possible outcome and their probability.

There are two types of probability distribution:

- Discrete Value, Probability Mass Distribution

  Ex. Dice number
- Continuous Value, Probability Density Distribution

  Ex. Income.

Some Terminology in Probability Distribution

From the definition we know that :

Probability distribution is a way to summarize every possible outcome and their probability.

Every possible outcome is quantified into numerical value that is called **random variable**.

ex. when we toss two coins. Here is the sample space :

(Face, Tail), (Tail, Face), (Face, Face), (Tail, Tail)

Let's say a random variable X = #numer of face occur. it will be 0 1 and 2

So, X is random variable with every possible values are 0, 1, or 2

Probability Mass Distribution Illustration

two-tosses	head-head	head-tail	tail-head	tail-tail
probability	0.25	0.25	0.25	0.25

This is the Probability Mass Distribution

X = #numer of head occur

0, 1, 2 (discrete value)

Rules:

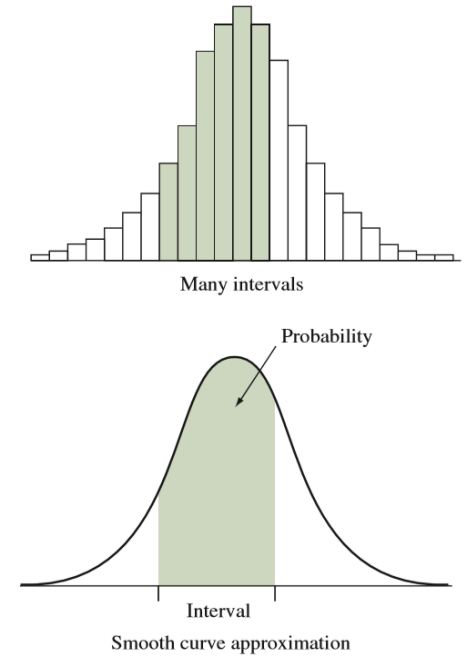
1. All add up to 1
2. Must be disjoint (mutually exclusive)
3. Each should be between 0 and 1

X	2	1	0
probability	0.25	0.5	0.25

Probability Density Distribution Illustration

Probability distribution is also a way to summarize every possible outcome and their probability when the **random variable** is **continuous**.

For example, probability to measure someone height exactly at 175 cm (no more or less than few picometer) is 0 or at least close to 0.



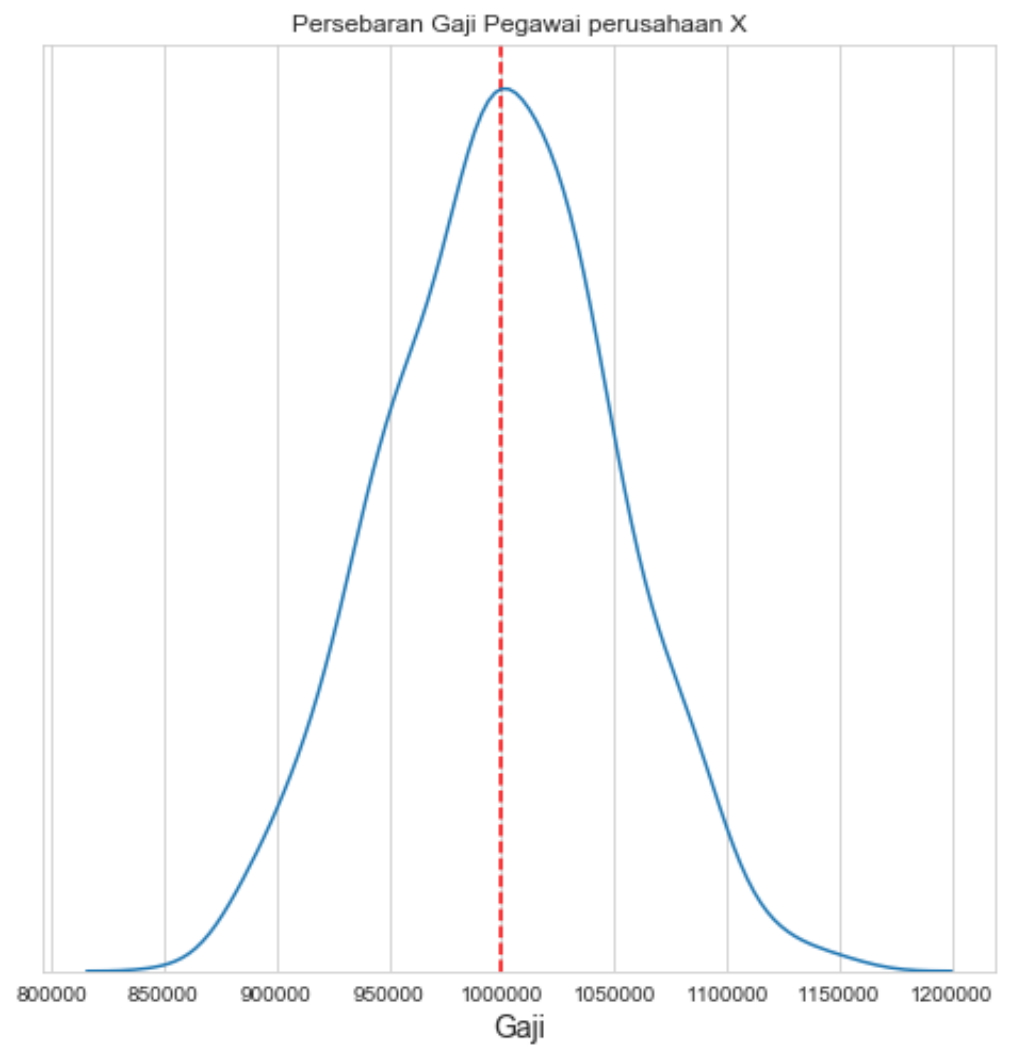
So, are measured based on the value range, not just one value. In the left picture we divide height into so many interval and compute the probability in each interval (similar like histogram right?)

In the next picture, it is an illustration of the smooth version of the histogram (infinite interval) which is drawn using a curve.

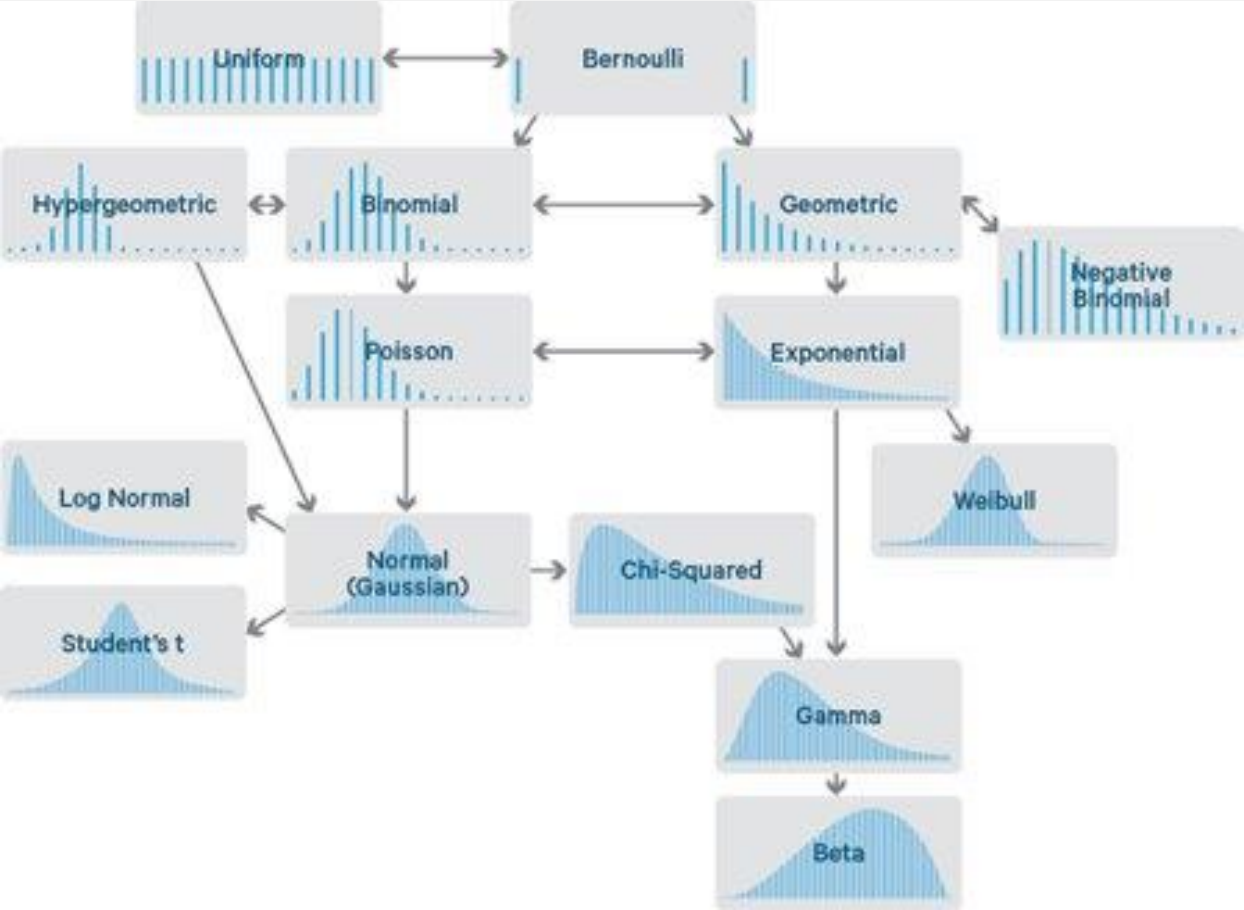
- each interval 175 - 180, 180 - 185 has a value of probability between 0 and 1
- an interval contain all possible value has a value of probability equal to 1.

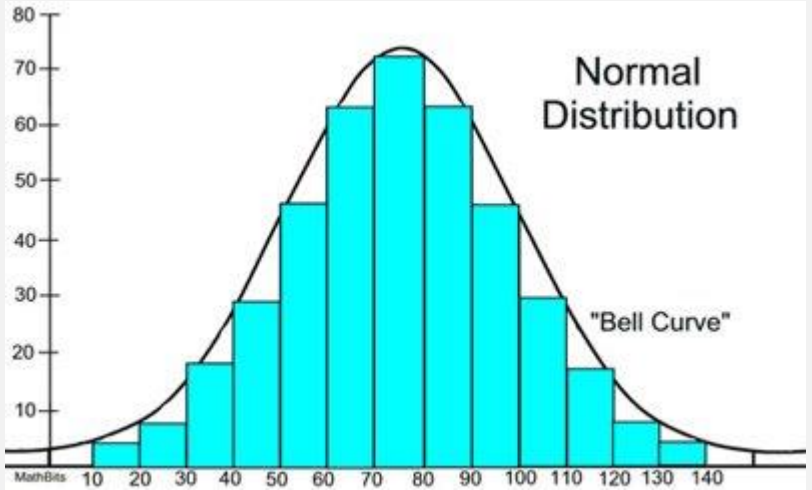
Probability Distribution

- *Probability Distribution* is useful for inferring the most probable event to be happen, likelihood of an event to occur, and the likelihood interval for event to occur. For example, company X employ 1000 employees and I want to take a sample of the salary for each employee. If I create the distribution plot of the company X employees salary, it could be visualized in the image beside.
- Picture beside giving us the *Probability Distribution* of the salary likelihood in the company X. We know if we randomly take a salary sample from company X, we probably would get a higher chance to acquire value that close to the red line or the mean. In other hand, the probability for us to get the random sample with salary value less than 900000 or more than 1100000 is small.



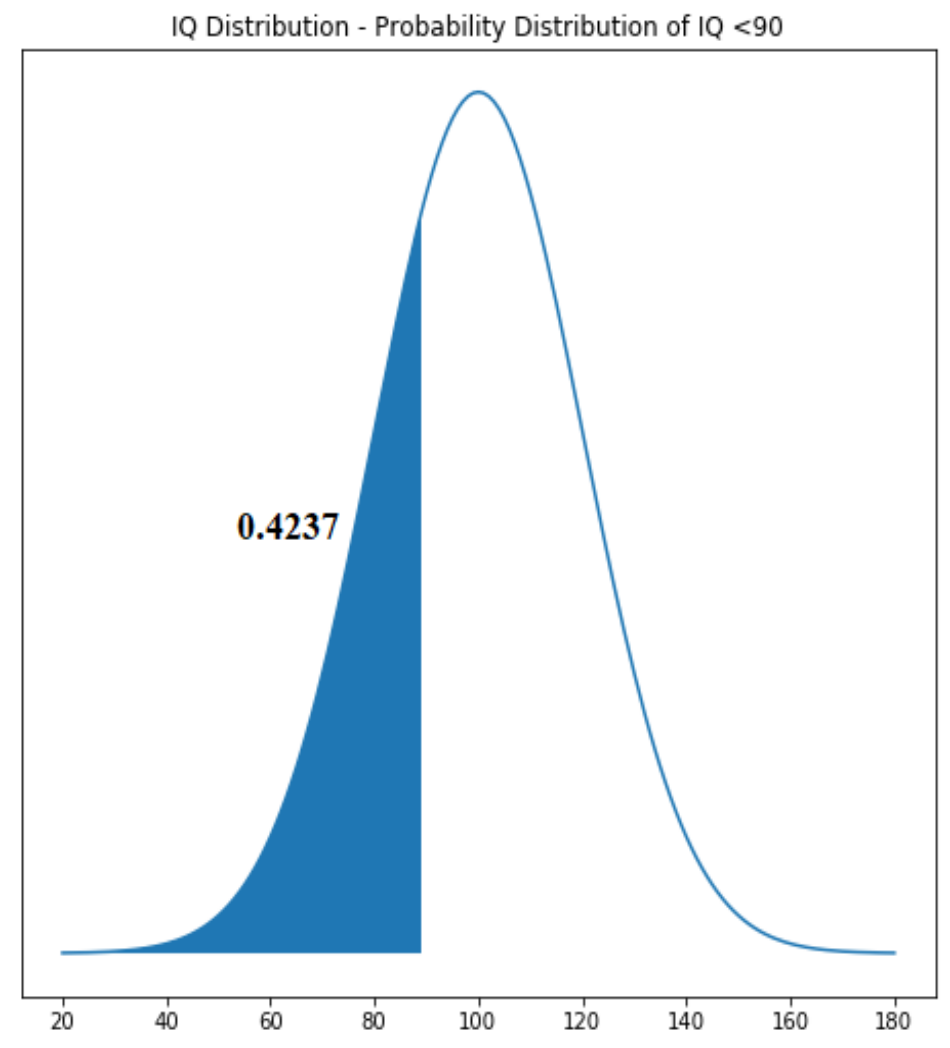
Statistical Distributions





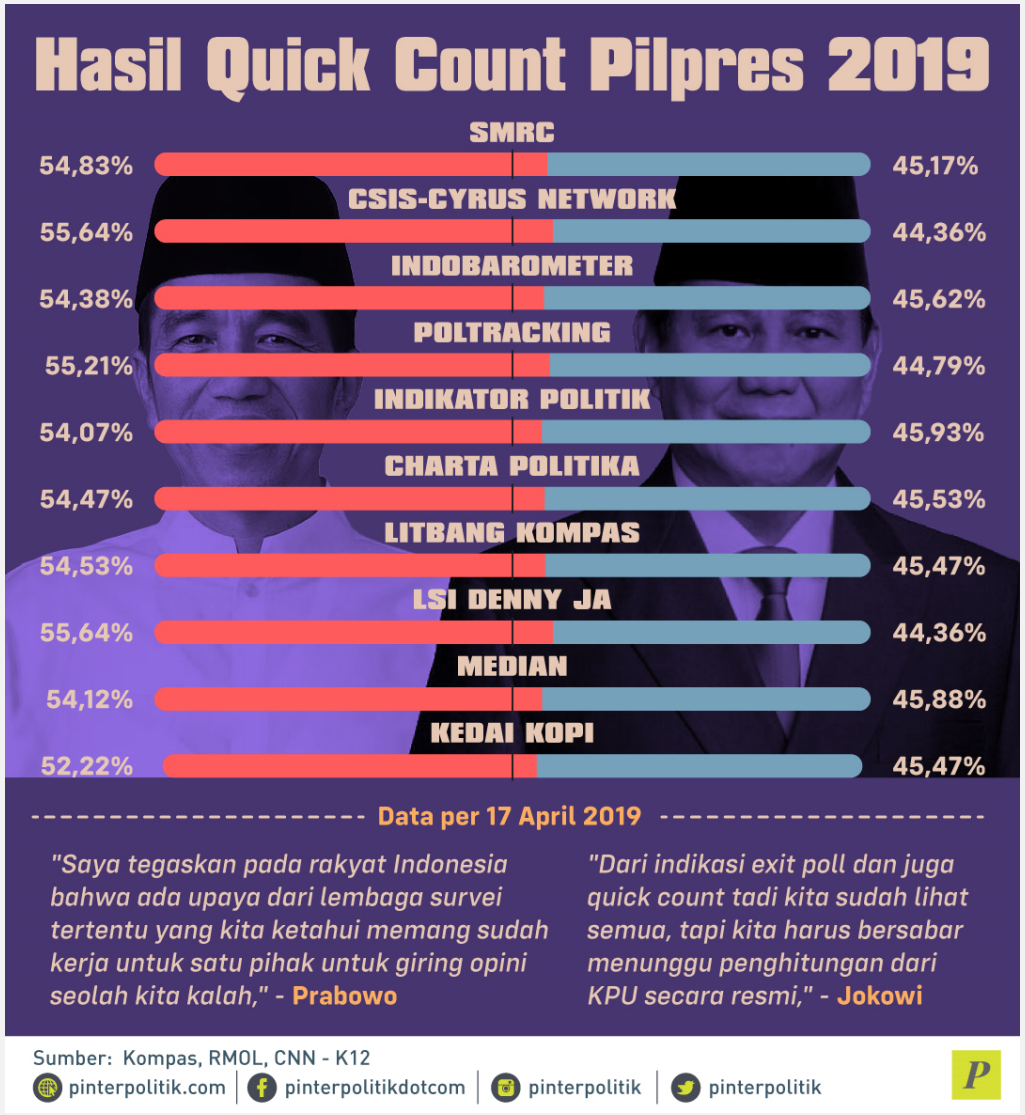
PDF

- In picture beside, we could for example want to know the probability to acquire a sample with IQ less than 90. We can measure the probability by measure the total area of the colored area or *area under the curve*. You could check this article to know more about the measurement. From the data, if we take a random sample, the probability to get a sample with IQ less than 90 is 42.37%.
- P(X < 90) = 0.4237
- X = random variable IQ



Sampling Distribution

- estimation from quick count is done based on random sample.
- estimation result vary from sample to sample.
- We can use histogram to infer the shape distribution of estimation result.
- The shape of the distribution estimation result is called sampling distribution.



Central Limit Theorem

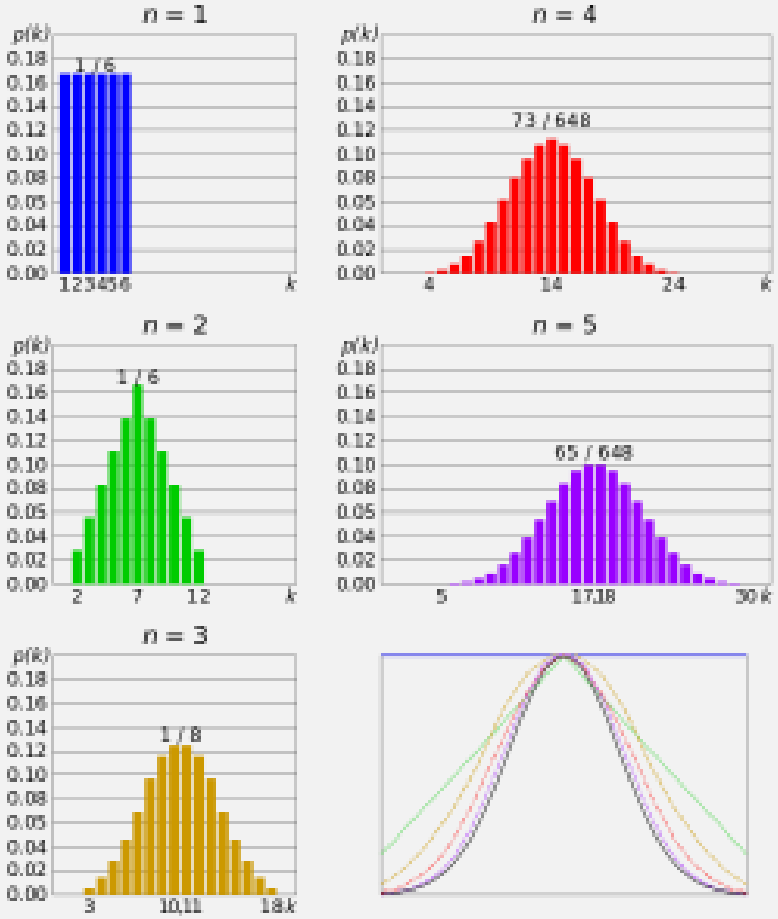
In the illustration before, we that sample taken 10 times and we have 10 estimation result.

For each sample, if **sample size large** enough (n > 30). **Distribution of the estimation** result from each sample should follow **normal** distribution.

This law also applied:

- to any statistics like mean, sum, median, etc (in this case proportion).
- when the population distribution is not normal.

Here's what the Central Limit Theorem is saying, graphically. The picture below shows one of the simplest types of test: rolling a die. The **more times you roll the die**, the more likely the shape of the distribution of the means tends to look like a **normal distribution graph**.

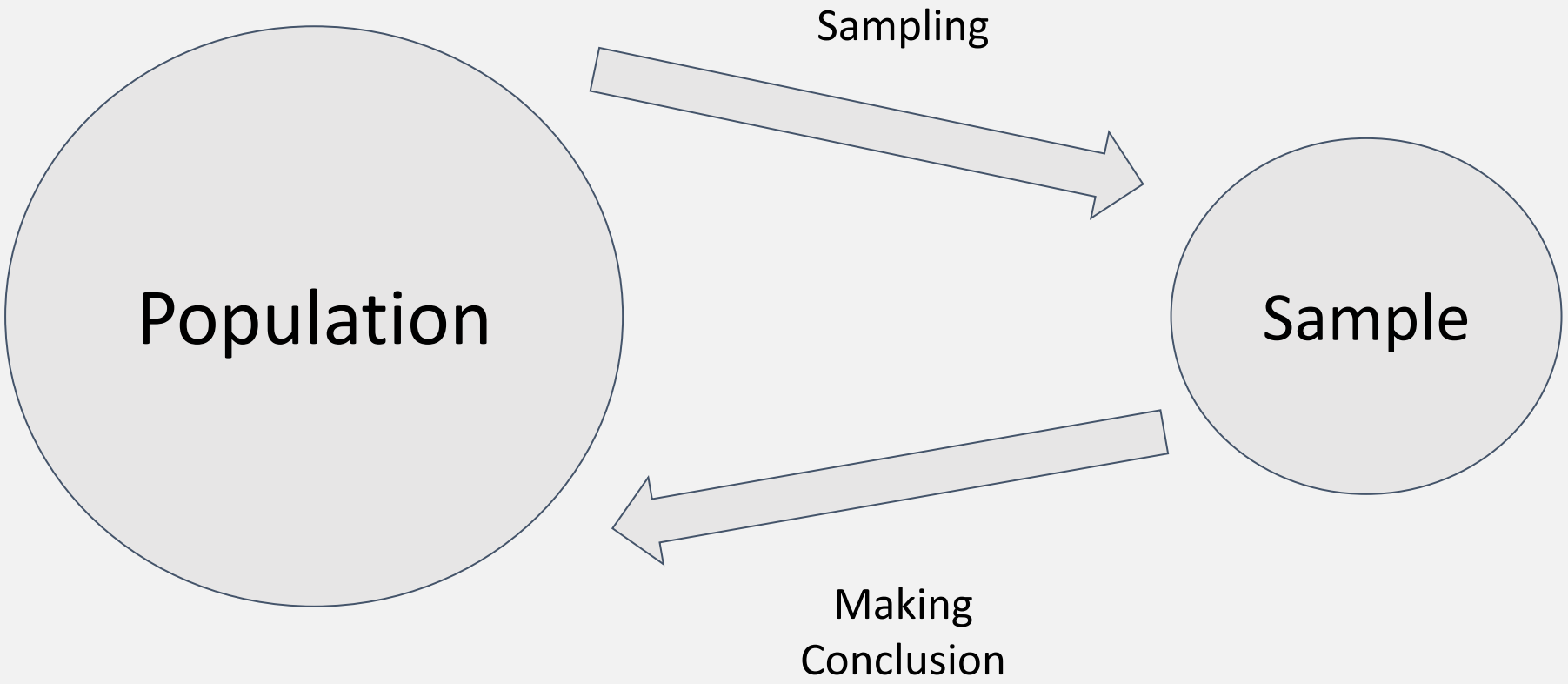


Why is Central Limit Theorem is so important ?

- Hypothesis testing like t-test, z-test, Anova F-test each of them need normality assumption to infer any population parameter using statistics
- With central limit theorem we know that any statistics if the **sample size large enough** the sampling distribution will follow normal distribution
- So, if we do any test statistics (t-test, z-test, F-test) and the **sample size** is **large** enough the result will be valid.

Inferential Statistics

Inferential Statistics



Inferential Statistics

- Inferential statistics uses **samples** to **generalizations** any phenomenon in the **population**.
- Inferential statistics uses **probability** to measure the **reliability of conclusion**.
- Method usually used to do inference are:
  - point estimation
  - interval estimation
  - hypothesis testing

Hypothesis Testing

- What is hypothesis
- Hypothesis testing component
  - Hypothesis
  - Assumption
  - Test Statistics
  - P-value
  - Conclusion

Components in Hypothesis Testing

- Hypothesis
- Assumption
- Test Statistics
- P-value
- Draw Conclusion

Illustration

When tossing a coin, we have **presumption of innocent** that the probability of each side would be 0.5 (face-tail).

- scenario 1, if we throw 20 times, is it still possible face show 9 times ?
- scenario 1, if we throw 20 times and we repeat it let's say 1000 times, is it still acceptable to say that the probability is 0.5 if we often get face < 7, let's say 800 of 1000 trials.
- how do you quantifying each scenario into numerical value?
- can we prove our presumption of innocent ?

What is Hypothesis ?

A **Hypothesis** is a **statement about population**.

Ex. we have a hypothesis that each probability for each side of the coin is 0.5. Ho : **P** = 0.5 (the coin is balanced) vs Ha : **P** ≠ 0.5 (the coin is not balanced)

A Hypothesis are consist of Null Hypothesis (Ho) and Alternative Hypothesis (Ha).

Characteristic of Ho:

- Ho is **presumption of innocent**.
- Usually contain a particular value.

  ex. Ho : **μ** = 50, Ho : **P** ≥ 0.5
- Usually takes no effect.

  ex. Ho : Beta = 0, Ho : dice is balance

Characteristic of Ha:

- Usually takes no certain value/ interval.

  ex. Ha : **μ** ≠ 50, Ha : **P** < 0.5
- Usually takes any effect.

  ex. Ha : Beta < 0, Ha : Beta > 0, Ha : dice is not balance.

P.I.C.O.T

Population, **I**ntervention, **C**omparison, **O**utcome, **T**ime

Good Example

- **H0**: Amazon.com visitors that receive Layout B will not have higher end-of-visit conversion rates compares to visitors that receive Layout A
- **H1**: Amazon.com visitors that receive Layout B will have higher end-of-visit conversion rates compared to visitors that receive layout A

Bad Example

- **H0:** Banks with nicer colors will not affect loan repayment
- **H1:** Banks with nicer colors will affect loan repayment

- **H0**: Amazon.com visitors that receive Layout B will not have higher end-of-visit conversion rates compares to visitors that receive Layout A
- **H1**: Amazon.com visitors that receive Layout B will have higher end-of-visit conversion rates compared to visitors that receive layout A

Population : visitors of Amazon.com
Intervention : new layout (layout B)
Comparison : visitors receiving Layout A
Outcome : Conversion rate
Time : End of visit to Amazon.com

- **H0:** Banks with nicer colors will not affect loan repayment
- **H1:** Banks with nicer colors will affect loan repayment

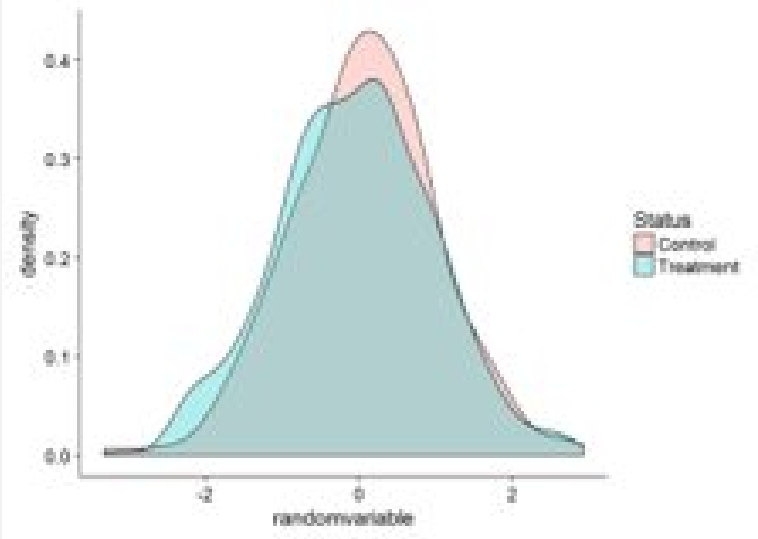
Poor Population definition : What banks, where and what level ? bank at certain city
Poor **I**ntervention definition : no clear definition of nicer colors
Poor Outcome definition : what effect are you measured, loan default rates, days past due, total branch loss ?

Measurements

CATEGORICAL VARIABLE

Conversion rate or proportion / sum or count

NUMERICAL VARIABLE



Distribution (mean, median, percentile) / difference in ratios

Randomness

Let's analyze data gathered from a hospital.

	Heart Disease	No Heart Disease	Total
Age > 50	57.8 %	42.2 %	100%
Age < 50	88.8 %	11.2 %	100%

We are talking about this example again

- Notice that young people have higher proportion suffer from heart disease 88.8%.
- Is that make sense that younger people have tend to have higher risk of suffering from heart disease?

This data is not representative and not gathered randomly:

1. This sample is **biased**
2. There is an unmeasured variable (**covariates**) which is self awareness for young people lower than old people
3. self awareness (covariate) affect the outcome not the one we want to know (Young/Old)

Which one is Random?

1. Participants are allowed to decide whether to be in a treatment or control group
2. Any participant with a national ID number ending in an odd number is assigned to treatment, any participant with an ID ending in an even number is assigned to control.
3. Participants east of my office are assigned to control, participants west of my office are assigned to treatment
4. We flip a coin to decide whether a participant is control or treatment

Which Method To Use ?

Different hypothesis testing usually refer to

- Type of Data used
- Sample Size
- Assumption
  - The form of population distribution
  - Variance
  - Sampling method
  - Randomization
  - etc

Different Assumption will leads to different method

Test Statistics

Test Statistics follow certain distribution and measure how far point estimate fall from the parameter in the Ho.

Test Statistics depend on what method we used to test the hypothesis

Some Popular test statistic:

- t score, used in t-test for mean population test
- z score, used in z-test for proportion population test (*also work for mean population test)
- F score, used in one way ANOVA when we test mean from > 2 population

In General Test Statistics are consist of ratio between magnitude and variability

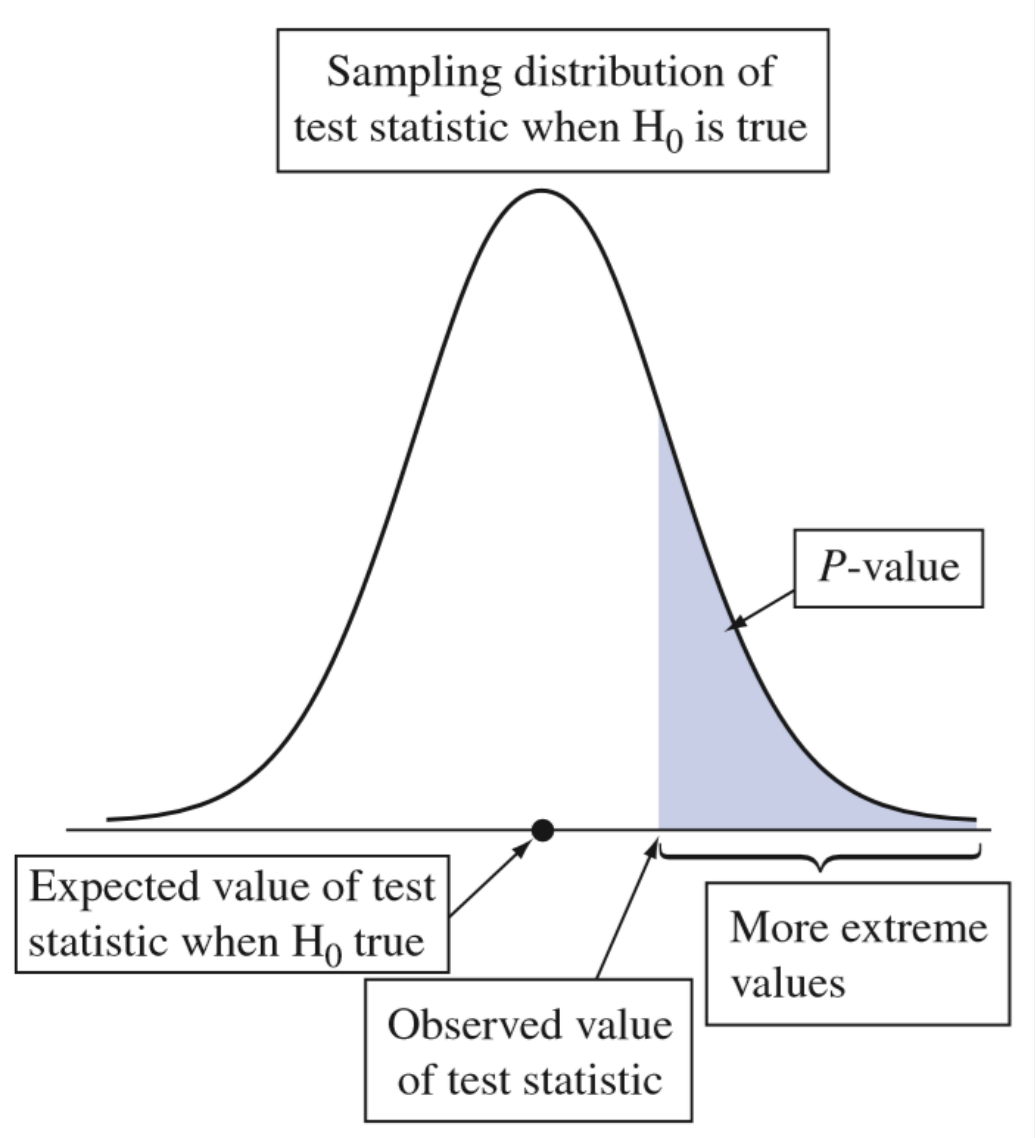
- magnitude (ex. mean different, proportion different)
- variability (ex. standard error)

P-Value

In Python and any statistical software, every test statistics will be converted into p-value (risk/probability). So, P-value is used to interpret any test statistics

P-Value is risk/probability to reject Ho (i.e. **μ** = 50) while the population actually Ho (i.e. **μ** = 50) based on sample data.

We reject Ho when P-value is equal to or below certain level of risk (Type of error I or **alpha**, α). Level of risk usually 0.1, 0.05, 0.01.



Drawing Conclusion

Final decision of hypothesis testing is whether we reject or not reject Ho.

- When we **reject Ho** we conclude that we have **enough evidence** that Ho is wrong below certain level of risk. P-Value and test statistic are the quantification of evidence.
- When we **do not reject Ho** it doesn't mean that Ho is true because **Ho** is **a presumption of innocent**. We just don't have **enough evidence** to proof that Ho is wrong.

Let's say we have hypothesis about mean population of Bogor citizen's age is below 33 or not.

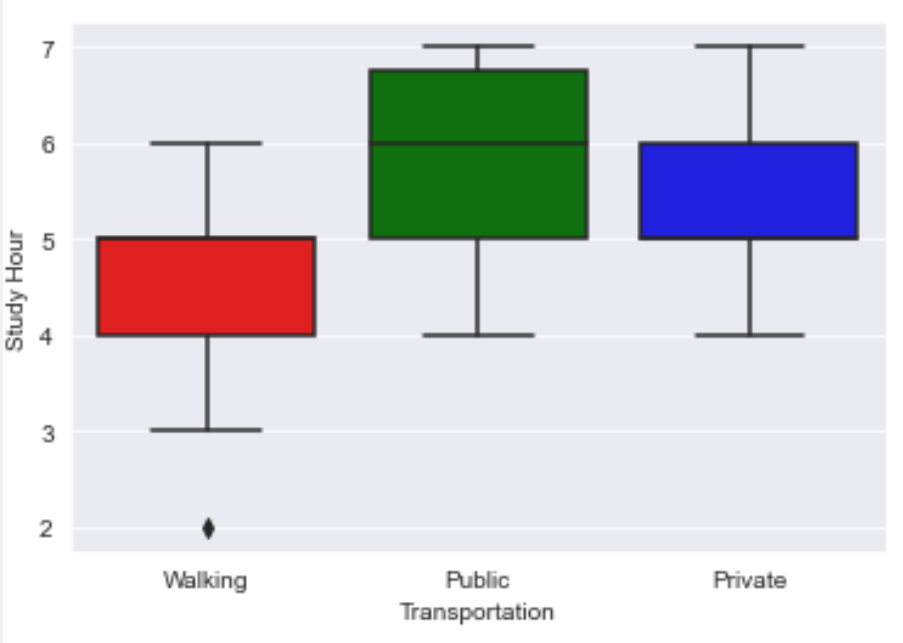
Ho : $\mu = 33$ vs Ha : $\mu < 33$

If we have P-value 0.0005. It means that if Ho is correct, it is very rare (0.0005) that if we get the sample we have.

If we have P-value 0.5. We accept Ho or fail to reject Ho. We can't say age mean population is equal to 33 Because we never know the actual value of age mean population and a hypothesis is a statement about population.

Why Do We Need Hypothesis Testing ?

For Example, we use hypothesis testing to compare mean between populations. Let's say we want to compare means of study hour between student who walking, using public transport, and using private transportation. here is the graphical summary.



The problem is **descriptive statistics** like this is **vulnerable to subjectivity**. I may say "no different" between student who use private and public. what about everyone else ?

Thank You!

PURWADHIKA HEAD OFFICE

BSD Green Office Park 9 - G Floor,
BSD City, Tangerang, Banten

CAMPUS BRANCHES

Purwadhika BSD
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